DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to VTU)

Department of Computer Science and Design

ECO TRACK

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USN's:

ECO TRACK

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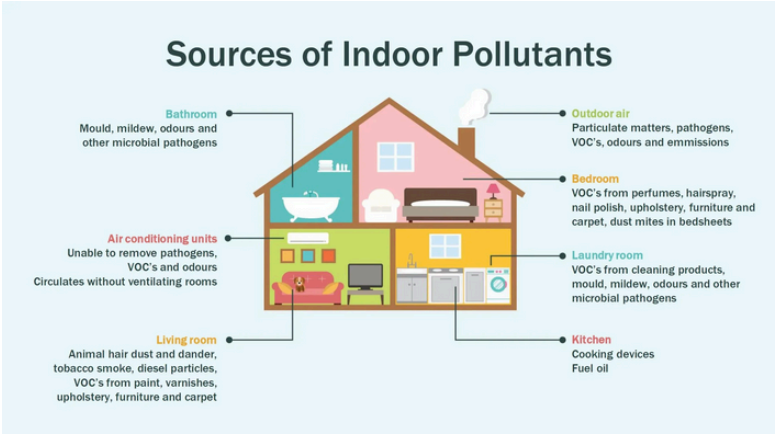
INTRODUCTION

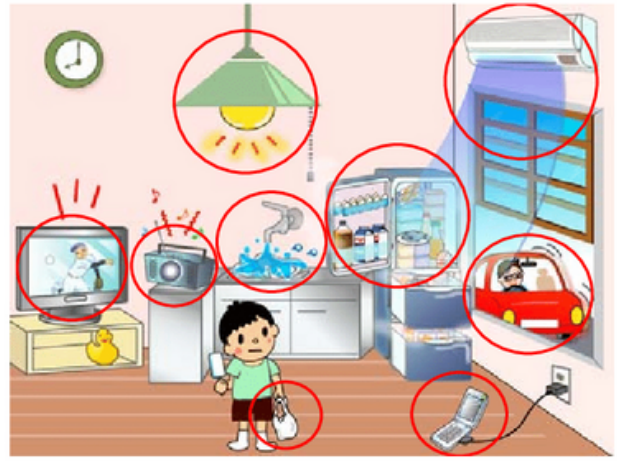
Our project also measures electrical usage and provides insights as to where electricity can be saved in our daily life usage.

We will also be measuring the CO2 emission and measuring the quality of air in a room or hall

PROBLEM IDENTIFICATION

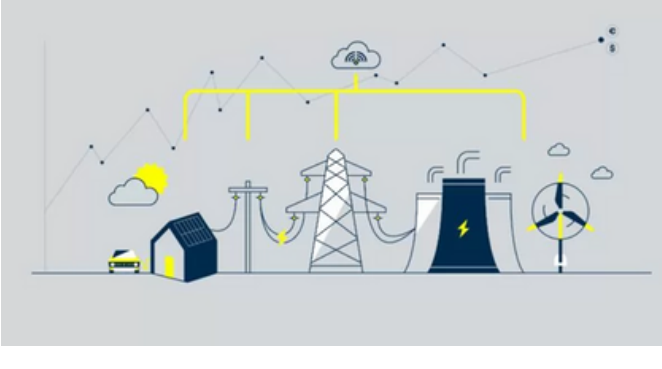
- Many users are unaware of how much electricity individual devices consume, leading to unnecessary power wastage.
- Existing systems mostly show total electricity usage and do not provide device-wise monitoring.
- Poor indoor air quality is often unnoticed, even though it directly affects health and productivity.
- Most AQI monitoring systems focus on outdoor air quality, ignoring indoor environments such as homes and classrooms.
- Electricity monitoring and AQI monitoring are usually implemented as separate systems.
- The lack of an integrated monitoring platform makes it difficult to analyze energy usage and air quality together.





LITERATURE SURVEY

- Electricity consumption has increased due to the extensive use of electronic devices in homes and campuses.
- Many users are unaware of device-wise power consumption, which leads to energy wastage.
- Existing energy monitoring systems mainly show total electricity usage and lack detailed visualization.
- Air pollution and poor indoor air quality have become serious health concerns in urban environments.
- Most AQI monitoring systems focus on outdoor air quality, while indoor monitoring is limited.
- Very few systems integrate electricity monitoring and AQI monitoring into a single platform.
- This creates a need for a simple, low-cost system to monitor electricity usage and indoor air quality together.



OBJECTIVE

Objective 1:
To measure and monitor indoor Air Quality Index (AQI) inside a household using an AQI sensor integrated with Raspberry Pi.

Objective 2:
To measure outdoor AQI levels on high-traffic roads and record the data daily for one week to study pollution variation.

Objective 3:
To measure the voltage at a socket when a charger is plugged in and switched ON without any device connected, in order to identify standby power conditions.

Objective 4:
To measure electricity parameters of common household appliances such as television, Wi-Fi router, refrigerator, and geyser when they are in use.

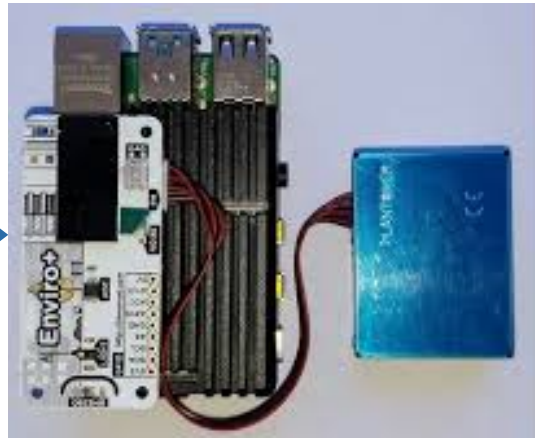
Objective 5:
To integrate AQI and electricity data into the EcoTrack dashboard for visualization and analysis.

Objective 6:
To increase awareness about energy wastage and air quality conditions through graphical representation.

Methodology

Step 1: AQI Data Collection

- An air quality sensor is connected to a Raspberry Pi.
- AQI readings are collected indoors (household) and outdoors (high-traffic roads).
- Outdoor AQI data is recorded daily for one week for comparison.

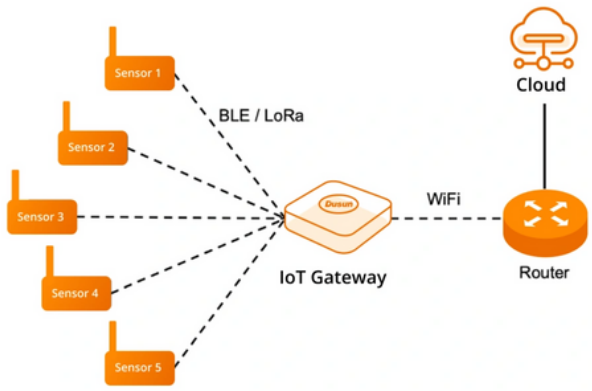


Step 2: Electricity Measurement

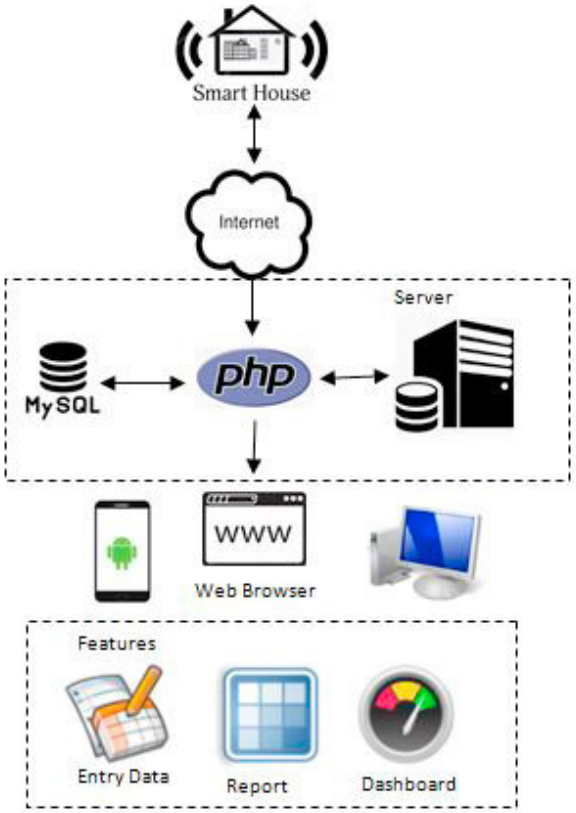
- Voltage is measured at the socket when a charger is plugged in and switched ON without a device.
- Electricity parameters of household appliances such as TV, Wi-Fi router, refrigerator, and geyser are recorded using smart plugs or energy meters.

Step 3: Data Transmission

- The Raspberry Pi sends AQI data to a remote Python backend server using the internet.
- Electricity data from smart plugs is also sent to the same server.
- All data is received through secure API endpoints.



Proposed System



Start

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AQI Sensor Measures Air Quality
(Indoor & Outdoor Locations)

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Raspberry Pi Collects AQI Data

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Electricity Data Collected
(Smart Plugs / Energy Meters)

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Data Sent to Python Backend Server
(via Internet / API)

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Backend Processes & Stores Data

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EcoTrack Dashboard Displays Data
(Graphs & Indicators)

↓

End

CONCLUSION

The EcoTrack system provides a simple and effective way to monitor electricity usage and air quality using sensor-based data collection and a centralized dashboard. By integrating AQI monitoring with electricity measurement, the system helps in understanding environmental conditions and identifying possible energy wastage. This project demonstrates how basic IoT and software technologies can be used to promote environmental awareness and sustainable practices.

THANK YOU!!